FLEET TEAM DIGEST

737MAX-FTD-34-20003

Issue Title : Honeywell Integrated Multi-Mode Receiver (IMMR) Dual GPS Dropouts

Airplane Model	ATA	Minor Model(s)
737MAX	3458-00	-8, -9
Other Model(s)	**ECCN**	
737NG, 777	9E991	

Originated Date	Last Revised Date	Created On
06/23/2020	03/18/2021	03/15/2021
Estimated Completion Date	**Next Update Date**	

Revision Description

Updated Status section to indicate completion of investigation and SB release.

Updated Final Actions section to include Boeing SB release information.

Updated Milestones section to provide latest release dates of the Boeing Service Bulletin.

Description

This FTD provides information regarding airplanes equipped with the IMMR Part Number, PN: 69002600-0101, experiencing dual GPS data being unavailable during the June 2020 timeframe.

Background

Boeing has received reports from multiple operators, of multiple airplanes experiencing events of GPS data becoming unavailable from both IMMRs. The GPS data recovers without performing any action, however, recovery time varied and lasted up to 3 hours.

When both IMMRs are unable to provide valid GPS data, GPS R INVALID and GPS L INVALID messages are annunciated on the FMS CDU, along with the illumination of the ATC FAIL light which is indicative of the ADS-B Out function failure.

In some cases, the outages occurred in areas of expected GPS outages per NOTAMs, but extended far beyond the geographic areas of the NOTAMs.

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Software version does not seem to be a factor in these events, since IMMRs with software PN HNR56-2601-0601 and IMMRs with the newer software PN HNR56-2601-0801 have both experienced the dual GPS outages.

The GPS outage can impact one or multiple IMMRs on a given airplane installation, can occur during any phase of flight, and typically impacts multiple co-located airplanes simultaneously within a given region.

Status

Honeywell has isolated root cause to an issue related to a specific monitor in the GNSS software. The monitor is timing out due to a signal being received from a satellite. The satellite triggering the issue is a test satellite which should not be accepted by the IMMR as a valid satellite. The test satellite serial number is being passed through a primary range check in the software, however, this is being flagged as out of bounds by the downstream monitor which is then timing out and effectively suspending GPS computations.

After successfully recording and playing back the satellite transmissions that causes an event. Honeywell further determined that an event will occur if both of the following conditions are met simultaneously and the IMMR enters acquisition mode:

Condition 1: GPS satellite PRN 01 is not being actively tracked by the IMMR.

Condition 2: A GLONASS satellite with a (test) serial number in the range of 25-32 is broadcasting and being tracked by the IMMR.

The following will cause the GNSS receiver to enter Acquisition mode:

- The IMMR is powered on through airplane power up or by cycling the IMMR Circuit Breaker
- The IMMR degrades from Navigation mode either due to insufficient satellites in view or in-band interference
- The IMMR GNSS receiver performs a self-reset

During an event, the IMMR’s GNSS receiver experiences a time-out of a software monitor that considers the test satellite serial number out of range, if GPS satellite PRN 01 is not actively being tracked. The time-out then causes the IMMR’s GNSS Receiver to automatically perform a reset. During this reset, the IMMR will continue to transmit all ARINC labels without interruption, but the labels associated with GPS data will be tagged as invalid (NCD). Labels not associated with GPS information, such as ILS guidance, will continue to be updated and transmitted normally. After resetting, the IMMR’s GNSS receiver will then re-enter acquisition mode and attempt to acquire satellites normally. However, if the two conditions listed above remain true, the response sequence will repeat. Once the conditions are no longer satisfied, the IMMR’s GNSS receiver will complete acquisition and transition to navigation mode normally with no flight crew or maintenance action required.

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After evaluating against FAA safety criteria and further analysis per Boeing safety review process, it has been found that there are no safety concerns related to this matter. Software changes to eliminate this issue have been identified and validated by Honeywell.

Boeing and Honeywell have completed the investigation regarding this matter and released a SB to install an updated IMMR OPS that resolves this concern and provides additional in-service fixes, such as the 7 day continuous operation shutdown, the pseudo range and carrier phase monitor out of range impediments.

Interim Action

Boeing and Honeywell advise that replacing the IMMRs will not clear the subject condition. GPS operation and data should automatically recover after up to 3 hours.

Honeywell has stated that no further NVRAM data downloads are needed as root cause has been determined.

Honeywell has released a satellite prediction tool that calculates the orbits of the specific navigation satellites that are influencing the availability of the IMMR GPS solution. The tool is provided to help anticipate upcoming date/times where interruptions may be experienced, at a user specific location. The tool uses the most recent satellite orbit information to predict time and area of interruptions. Please note position and altitude coordinates input into the tool do not need to be precise to generate a useful prediction and environmental variations may influence the exact times when outages begin and end. This tool can be used for flight planning scenarios, but is most useful in anticipating outages at a departure airport prior to aircraft dispatch and can be used by airline operators to predict: Probable Outage / Possible Outage / No Outage. Honeywell is currently investigating further improvements to allow for in route predictions.

Final Action

Boeing has released the Service Bulletin that allows on-wing installation of the updated IMMR software which resolves this matter.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Probable Root Cause Identified: 14-JUL-2020 (Complete)

Root Cause Established: 22-JUL-2020 (Complete)

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Honeywell TSO Approved: 18-NOV-2020 (Complete)

Boeing FAA Service Bulletin Available: SB737-34-3684 - 14-JAN-2021 (Complete)

Boeing EASA Service Bulletin Available: SB737-34-3684 Rev 1 - 10-MAR-2021 (Complete)

Operator Action

Please contact Honeywell for further information and access to the satellite prediction tool.

Currently, Honeywell believes it has obtained enough information and no longer needs operators to send in extracted data from the IMMR, but for tracking purposes, requests operators send a brief description when an event is encountered to Aerotechsupport@honeywell.com with the information details noted below. Additionally, Boeing also requests a copy of the information sent via a BCS message. At a minimum to investigate this issue, the following information would be needed: UTC date/time and location of event occurrence, and the ending UTC time/date of event for the GPS outage event being reported. Additionally, please list if there was action performed, if any, while on ground/in air to resolve the event.

Operator Name

Aircraft Type

Aircraft

IMMR P/N

IMMR S/N

IMMR OPS S/W P/N

GPS antenna P/N

Ground or Air

Fault Position (LH/RH/both)

GPS loss event start UTC Date & Time

GPS loss event stop UTC Date & Time

Problem reported

Cockpit effect

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Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	SB 737-34-3684	01	03/10/2021
Service Information Letter (SIL)	Honeywell #D202006001060	Rev.0	06/25/2020
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19001		
Service Bulletin (SB)	737-34-3483	NEW	10/21/2019
Fleet Team Digest Article (FTD)	737MAX-FTD-34-20001		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Integrated Multi-mode Receiver (IMMR)	69002600-0101	Honeywell International	

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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